Hierarchical information mediates time-course of access to items in linguistic memory

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Prominent models of real-time dependency formation posit a memory architecture that distinguishes between two representational states: *focus of attention* and *activated long-term memory* [1-2]. The capacity of the former is typically restricted to a single chunk that is immediately available [1-4]. Conversely, representations in the latter require explicit retrieval or reactivation [1-2,5-8]. Items within this store are distinguished via retrieval cues; crucially, no difference is predicted in their time course of retrieval, as all items are accessed in parallel. But what modulates which linguistic material is kept in active attention at any point in processing? We propose the maintenance of a linguistic item in active attention is determined by its hierarchical availability. We present novel evidence that hierarchically available *targets* are accessed significantly faster than unavailable *distractors*. This finding is consistent with the claim that linguistic representations can be accessed with a qualitatively different time course, suggesting a further functional divide within activated representations in linguistic memory, driven by hierarchical information.

Data: We employed a large-scale visual world eye-tracking study (N=152) to monitor the real-time access to referents in interpreting a Turkish local anaphor, a construction governed by hierarchical item-to-item relations (i.e., clause-mateness, c-command, co-argumenthood). In all sentences the reciprocal, *birbirleri,* was the object of an embedded clause, and the embedded subject served as a hierarchically available *target* (see (1)). We manipulated the hierarchical availability of a second referent, creating an ambiguous condition with a hierarchically available *target*, Indirect Object (IO), and another one with a hierarchically unavailable *distractor*. To isolate the role of hierarchical relations, both the IO and distractor were plural and marked with the same case, were clause-mates to the anaphor and appeared in the same linear position. Participants' task was to listen to the stimuli and click on the anaphor's referent on a visual display (**FIG1**).

Results: We used two methods of analyses to evaluate whether there was a difference in the time-course of looks to targets vs. distractors. Finding such a difference would indicate that differences in hierarchical availability correspond to different speeds of retrieval. Both analyses were restricted to trials with a subject response; this allowed us to compare looks to IO vs. distractor while holding the response constant. First, we ran Bayesian divergence point analyses on gaze proportions, which uses bootstrapping to estimate the time-point when looks to two referents start to significantly differ [9] (**FIG2**). We found strong evidence that the onset of divergence between the subject and IO (640ms [95%CrI: 575-705]) was later than that between the subject and distractor (190ms [176-204]) (BF_{10} > 100). But the onset of divergence between IO and unmentioned plural referent (313ms [263-364]) was earlier than that between the distractor and unmentioned referent (539ms [429-650]) (BF_{10} = 88.9). Together, these comparisons suggest that subject and IO were accessed earlier than the distractor. Second, we analyzed the time-course of new fixations [10], launched towards IO vs. distractor (**FIG3**). A Bayesian generalized additive mixed-effects model showed evidence for more new fixations towards IO than distractor within the 50-250ms window after the anaphor onset (all ms reports adjusted for 150 ms delay in saccade programming).

Overall, we find that hierarchically unavailable distractors are activated ~225ms later than targets. We take this to suggest that hierarchical positional information shapes the organization of linguistic items in memory. We propose that hierarchically prominent items within a clause (i.e., subjects and IOs) are kept in temporarily active, privileged memory addresses, potentially corresponding to the *focus of attention* or *region of direct access* in current domain-general theories of working memory [11-12]. Such an addressing scheme would facilitate the transient structural binding of content (i.e., linguistic items) to context (i.e., hierarchical position) [13], allowing rapid, direct access during retrieval. If hierarchically unavailable items lack this privileged representational state, then they must be retrieved through a slower, activation-driven mechanism. Methodologically, this work demonstrates that visual world data can be used to answer complex time-course questions.

Subject [Subject$_i$ IO$_k$ / Distractor$_l$ (Postposition) Adverb **Reciprocal** *Verb*] Verb

(1) Ayşe [kameraman-lar-ın$_i$ { yönetmen-ler-e$_k$ / yönetmen-ler-e$_l$ karşın }
A. cameraman-PL-GEN director-PL-DAT / director-PL-DAT despite
hevesle **birbirleri-ni$_{i/k/*l}$** *gösterdiklerini*] duydu.
enthusiastically each.other-ACC showed heard
'Ayşe heard that the cameramen$_i$ enthusiastically showed {*the directors$_k$ / despite the directors$_l$*} each other$_{i/k/*l}$.'

FIG1. Visual display with subject *cameramen* (top left), IO/distractor *directors* (top right), two unmentioned referents (bottom).

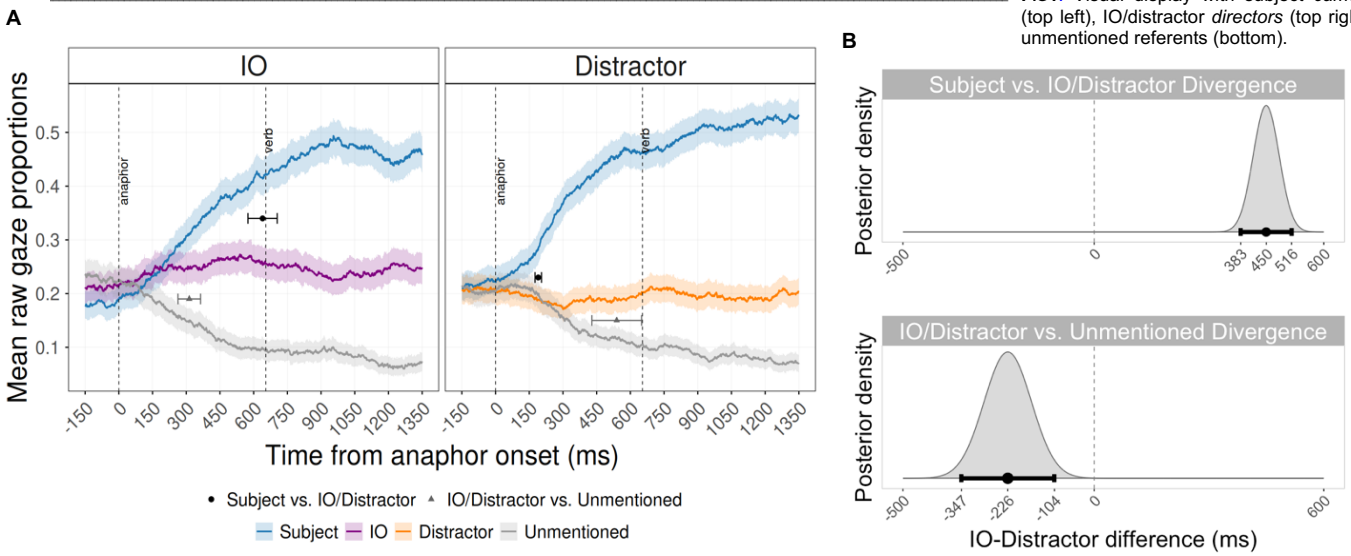

FIG2. **A.** Gaze proportions time-locked to the onset of the anaphor, extending 1500ms; with estimated onsets of divergence points for Subject vs. IO/Distractor, and IO/Distractor vs. Unmentioned (plural) looks. Fixation curves show mean gaze proportions, with error bands for bootstrapped 95% CrIs. The vertical dashed lines at 0 mark the 150-ms shifted onset of the anaphor and verb windows. Point estimates show means of divergence onsets in posterior distributions, with error bars for 95% CrIs. **B.** Posteriors of the difference between IO and Distractor conditions for the divergences between Subject vs. IO/Distractor, and IO/Distractor vs. Unmentioned, respectively. Point estimates show mean differences, with error bars for 95% CrIs. Vertical dashed lines mark a difference of 0.

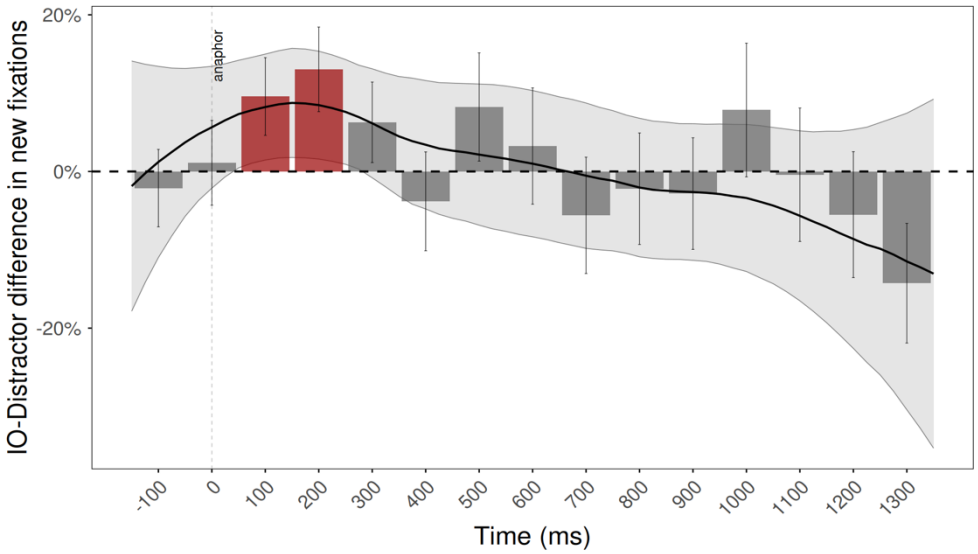

FIG3. Time course of the descriptive and predicted differences in new fixation proportions towards IO vs. Distractor, time-locked to the onset of the anaphor and extending 1500ms. The bar plot shows the descriptive mean differences calculated over 100-ms time bins, with error bars reflecting SEs of the mean differences. The overlaid solid line represents the predicted mean difference taken from the fitted Bayesian Generalized Additive Mixed Model (GAMM) spline function, while the shaded error bands reflect the 95% CrI of the predicted difference. The vertical dashed line at 0 marks the 150-ms shifted onset of the anaphor window, and the horizontal dashed line marks a difference of 0. Time bins highlighted in red correspond to the regions where the model's spline estimate suggests a reliable difference.

REFERENCES. [1] McElree, 2006 [2] Lewis & Vasishth, 2005 [3] Oberauer & Bialkova, 2009 [4] Wagers & McElree, 2022 [5] McElree et al., 2003 [6] Nicenboim & Vasishth, 2018 [7] Lisson et al., 2023 [8] Kush et al., 2015 [9] Stone et al., 2021 [10] Kingston et al., 2016 [11] Oberauer, 2002 [12] Oberauer, 2019 [13] Keshev et al., 2025